# Multivariate Decidable Fragments of Canonical Comparison

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**Abstract**

The canonical ordinal assignment used by Wilken and Weiermann turns a termination question for Gödel's $T$ into universal comparison of numerical collapse expressions. Several variables create a difficulty absent from closed evaluation: derived values share inputs and cannot be replaced by independent unknowns. We prove decision results which retain those correlations. For the exact control $\Phi(n) = F_5(n + 100)$, we verify the hypotheses of Snyatkov's published arithmetic theorem. This settles a finite-coefficient source class containing every recursor-free term with ground free variables, regardless of its internal types. We then give exact finite-degree descent kernels, ranked cancellation, and a terminating closed existential procedure. These decide canonical comparison when every internal type has level at most two. At arbitrary fixed ordinal heads, different arguments give separate decision results for nested common-head sums, balanced heterogeneous layers, and rows depending on one arbitrary positive core. The last result reduces directly to a finite Presburger predicate. All algorithms include their finite exceptions. None decides unrestricted comparison of all ground-open canonical assignments.

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# 1 What is being compared?

Wilken and Weiermann's assignment associates with a typed term a vector of ordinal expressions. Their effectiveness question asks, among other things, whether its universal comparison relations can be decided effectively [1]. The assignment is computed from the original term, so two terms with the same normal form can have different assigned vectors. We keep that canonical assignment throughout; normalization is not a preliminary step.

This paper concerns terms whose free variables have ground type. For their *actual generated assignments*, the positive coordinates are closed ordinals. The zeroth coordinate is a natural-valued expression made from positive-natural ground variables, natural constants, addition, fixed natural multiples, and functions

$$h_\lambda(n) = \psi(\lambda + n), \qquad 0 < \lambda < \varepsilon_0 \text{ a fixed limit.}$$

The forward-translation theorem in the foundations companion [3] proves this by simultaneous induction on generated terms and abstraction operations. The qualifier "actual generated" matters: the unrestricted ambient classes of ordinal expressions in the original definition also permit promotions of whole scalar expressions. Those larger classes do not have the same closure invariant merely by membership. We use the stronger proved invariant only for the source assignments to which it applies.

Use the additive Cantor-normal-form norm

$$N(0) = 0, \qquad N\left(\sum_i \omega^{\alpha_i} c_i\right) = \sum_i c_i\big(1 + N(\alpha_i)\big).$$

For a fixed limit set

$$J_\lambda(t) = \psi\big(\lambda + t - N(\lambda)\big), \qquad t \geq N(\lambda), \qquad J_0(t) = t. \tag{1.1}$$

Thus $t$ is the whole ordinal's norm, not just its finite tail. In particular, an offset $N(\lambda)$ must not disappear when two heads are compared. The foundations companion proves the exact closed evaluator, controlled strictness, and the complete-count identity

$$J_\lambda(t) = Z + \tau, \qquad Z = f^\tau(t), \qquad \tau \geq 1, \tag{1.2}$$

where $f = \Phi$ and $\tau$ counts replacements until the ordinal becomes finite. Finite successors after that point are not included in $\tau$. These are the foundational facts imported here, with their proofs in the included manuscript. The remaining comparison mechanisms are proved below.

Our results are complementary. A bound on internal type levels permits arbitrary arrangements of polynomial heads. Common-head and balanced certificates instead permit nonpolynomial heads but restrict how the open arguments are arranged. Neither condition subsumes the other. Failure of a certificate means that its theorem does not apply; it is never a negative answer to the requested comparison.

## 2 The exact control and a published arithmetic theorem

Define, literally,

$$F_0(n) = 2^n, \qquad F_{j+1}(n) = F_j^{n+1}(n), \qquad f(n) = F_5(n+100). \tag{2.1}$$

A superscript on a function denotes a finite iterate. The starting argument in $F_{j+1}$ varies with $n$. This convention will not be replaced by iteration from a fixed seed.

**Lemma 2.1** (Growth)**.** *Each $F_j$ is recursive and strictly increasing, with $F_j(n) \geq 2^n$. For $j \geq 0$,*

$$F_{j+1}(n+1) \geq 2^{F_{j+1}(n)+1}.$$

*Consequently $f(z+1) \geq (A+2)f(z)$ whenever $z \geq A \geq 0$.*

*Proof.* Finite iteration preserves recursiveness, strict increase, and domination of $2^n$. Increasing the input from $n$ to $n+1$ makes the first $n+1$ iterates at least one larger, and the definition then supplies one further application of $F_j$. Thus $F_{j+1}(n+1) \geq F_j(F_{j+1}(n)+1) \geq 2^{F_{j+1}(n)+1}$. For $v \geq 1$, $2^{v+1} \geq (v+2)v$, by direct checking at $v = 1, 2, 3$ and induction. Apply this with $v = f(z) > z \geq A$. □

**Lemma 2.2** (Effective constant residues)**.** *For every $m \geq 1$ one can compute $r_m \in [0, m)$ such that*

$$F_j(n) \equiv r_m \pmod{m} \quad (j \geq 1,\ n \geq m).$$

*In particular $f(n) \equiv r_m \pmod{m}$ for $n \geq \max(0, m-100)$. For $s \geq 1$, $j \geq 2$, this can be strengthened to*

$$m \leq 2^{2^s} \quad \Longrightarrow \quad F_j(s) \equiv r_m \pmod{m}. \tag{2.2}$$

*If $B = f(u)$ and $m \leq 2^{2^B}$, then also $f(u+1) \equiv r_m \pmod{m}$.*

*Proof.* Set $r_1 = 0$. For $m > 1$, write $m = 2^a b$ with $b$ odd, and put $q = \varphi(b)$ if $b > 1$, $q = 1$ otherwise. Since $q < m$, recursively compute $r_q$, then use the Chinese remainder theorem to define

$$r_m \equiv 0 \pmod{2^a}, \qquad r_m \equiv 2^{r_q} \pmod{b}.$$

The recursion strictly lowers the integer modulus. For towers $T_0(n) = n$, $T_{h+1}(n) = 2^{T_h(n)}$, induction on $m$ shows $T_h(n) \equiv r_m \pmod{m}$ whenever $h, n \geq m$. Indeed, the last exponent is at least $n \geq a$, while its residue modulo $q$ is $r_q$ by induction; Euler's theorem handles the odd factor. This proves the assertion for $F_1(n) = T_{n+1}(n)$. At every higher level, all intermediate arguments are at least $n$, so the last application of the preceding level proves the same residue assertion.

For the stronger bound, $F_1(s) \geq 2^{2^s}$. In $F_2(s)$ there are at least two $F_1$ applications; the final one has input at least $F_1(s)$, so its input is at least $m$. The same argument propagates through every $j \geq 2$. Finally put $n = u + 100$. The last $F_4$ call in $F_5(n+1) = F_4^{n+2}(n+1)$ has input $F_4^{n+1}(n+1) > F_4^{n+1}(n) = B$. Apply the stronger bound to that call. □

**Theorem 2.3** (Snyatkov's arithmetic supplier)**.** *The first-order theory* $\mathrm{Th}(\mathbb{N}; 0, 1, +, <, f)$*, with the exact function in* (2.1)*, is decidable.*

*Proof.* We use Theorem 2 and effective Corollary 2 of Snyatkov [2], pp. 50–51. Its definitions on p. 40 require an increasing recursive function effectively compatible with addition: effective eventual periodicity modulo every integer, and an effective shifted sign separation for each unbounded integer combination of fixed translates. It also requires effective eventual periodicity of the *fixed-seed* hyperfunction $H(0) = 1$, $H(n+1) = f(H(n))$. The cited theorem gives a decision procedure in an expanded language with the stated auxiliary divisibility and inverse symbols, hence in its $f$-only reduct.

The residue condition follows from Lemma 2.2. For the second condition combine equal shifts in $S(x) = \sum_b a_b f(x+b)$. If every combined coefficient is zero the sum is zero. Otherwise let $B$ be the largest shift with nonzero coefficient, and set

$$A = \sum_{b<B} |a_b|, \qquad \delta = \max(0, -\min b, A + 1 - B).$$

Every argument at $x + \delta$ is nonnegative. With $z = x + \delta + B - 1 \geq x + A$, Lemma 2.1 gives, if $a_B > 0$,

$$S(x+\delta) \geq f(z+1) - Af(z) \geq 2f(z) > f(x).$$

For $a_B < 0$, apply the same calculation to $-S$. This proves unboundedness, its sign, and a suitable shift by finite arithmetic on the input data. Lastly $H(n) \geq n+1$; for $n \geq m$, the last application of $f$ in $H(n)$ has input at least $m$, so $H(n) \equiv r_m \pmod{m}$. All published hypotheses are therefore paid. No identification of $H$ with a diagonal $F_j$, and no decision result for variable-seed iteration, is being imported. □

## 3 An actual-source finite-coefficient class

The foundations companion proves the identity

$$\psi(\omega k + n) = f^k(n + 2k) + k \qquad (k, n \in \mathbb{N}). \tag{3.1}$$

Although an iteration occurs here, its length $k$ is a closed integer computed from source syntax. Replacing the expression by $k$ copies of $f$ is an effective finite translation.

**Theorem 3.1** (Finite-coefficient source comparison)**.** *Take two original well-typed $T$ terms of the same type with ground free variables. Compute their canonical assignments. Suppose that at every zeroth-coordinate collapse $\psi(\omega \otimes c + g)$ its closed coefficient $c$ has finite ordinal value. Their universal weak, WW-strict, and equality comparisons are decidable. Every recursor-free such term passes this test, with no bound on its internal types.*

*Proof.* Use the actual-source forward translation of [3]; $\otimes$ is natural product at this source interface. Evaluate closed positive coordinates first. Weak vector comparison requires weakness in all coordinates; WW-strict comparison requires strictness at coordinate zero and weakness above it. Equality requires equality everywhere. A failed positive-coordinate test already decides the requested universal relation.

For the remaining scalar expressions, retain each original variable name, addition, and each fixed coefficient. Replace a collapse with finite coefficient $k$ by $g \mapsto f^k(g + 2k) + k$ using (3.1). This translation preserves its value at every positive valuation. A ground vector has only one coordinate, so admissible closed substitutions give exactly all positive integers; there is no hidden

coupling between distinct ground variables. Apply Theorem 2.3 to the required universal formula, with $x_i \geq 1$ explicitly imposed.

The source theorem also proves that recursor-free actual assignments have finite positive coefficients, by induction on their term and abstraction construction. Thus all these sources pass the test. This argument uses that actual-source theorem, not an assertion about all ambient ordinal-expression classes. A recursor need not cause failure: the canonical vector of $R^o X$ is $(f(x+4)+1, 1, \omega)$, whose direct scalar collapse coefficient is one. Closed coefficient evaluation and the finite syntax traversal terminate. □

## 4 Exact finite-degree prefix kernels

The next class requires counters as well as terminal norms. A dominant approximation would lose cancellations needed for multivariate comparison. For integer $d \geq 1$, retained norm $c \geq 0$, and sufficiently large $t$, start with suffix $\omega^d$. At step $i \geq 1$ use total norm $f^i(t)$ and replace the nonfinite suffix by the greatest smaller suffix of norm at most $f^i(t) - c$. Stop as soon as it is finite. Let $E^d_c(t)$ be the number of steps and

$$X^d_c(t) = f^{E^d_c(t)}(t).$$

Each replacement strictly decreases an ordinal, so the process terminates. The exact maximal-predecessor theorem in [3] supplies the finite algorithm and full-budget saturation. The numerical definition makes sense even for $c$ not realized by any actual ordinal prefix.

For a genuine prefix $\mu$ whose least CNF exponent is at least $d$, and $c = N(\mu)$, splitting the controlled history gives

$$J_{\mu+\omega^d}(t) = J_\mu(X^d_c(t)) + E^d_c(t), \qquad t \geq c + d + 1. \tag{4.1}$$

The output of $X$ is still the *total* norm, so the prefix mass is not subtracted again. For fixed signed $j$, put $X^d_{c,j}(t) = f^{E^d_c(t)+j}(t)$ on a tail where the exponent is positive. These are coherent depths: $f(X^d_{c,j}) = X^d_{c,j+1}$. A negative $j$ is a fixed number of already available predecessors under $f$; it is not a new variable-iteration graph.

**Lemma 4.1** (Exact block recursion)**.** *One has*

$$E^1_c = 1, \quad X^1_c = f, \qquad E^2_c(t) = f(t)/2 - \lceil c/2 \rceil + 1.$$

*Define*

$$V_j(t) = f^{\,f(t)/2+j+1}(t). \tag{4.2}$$

*Then* $X^2_{c,j} = V_{j-\lceil c/2 \rceil}$. *For* $d \geq 2$, *on a computable tail,*

$$X^d_{c,j} = X^{d-1}_{c,j} \circ X^d_{c+d}, \tag{4.3}$$

$$E^d_c = E^{d-1}_c \circ X^d_{c+d} + E^d_{c+d}. \tag{4.4}$$

*Proof.* For $d = 1$ the first predecessor is finite. For $d = 2$, the first predecessor has $\lfloor (f(t) - c)/2 \rfloor$ copies of $\omega$ and a finite remainder; each $\omega$ copy then costs one replacement. The function $f$ is even, which gives the displayed formula. For general $d$ write $B = f(t)$ and $B - c = dq + \rho$, $0 \leq \rho < d$. The first predecessor has leading part $\omega^{d-1} q$. Its remainder is zero if $\rho = 0$, finite one

if $\rho = 1$, and $\omega^{\rho-1}$ if $\rho \geq 2$. This is the largest remainder *below* $\omega^{d-1}$ under the norm budget, not a maximum with no ordinal bound. Removing it leaves norm $B$ with spent count one in the first two cases, or norm $X^{\rho-1}_{B-\rho}(B)$ with count $1 + E^{\rho-1}_{B-\rho}(B)$ otherwise. The remaining $q$ degree-$(d-1)$ blocks have retained masses $c + d(k-1)$, for $k = q, q-1, \ldots, 1$. Their $X$ maps compose and their $E$ counts add. Replacing $c$ by $c + d$ preserves the initial remainder and initial entry but stops one block earlier. This proves both recurrences, including the coherent final depth $j$. It is also a rank-recursive exact evaluator for every displayed kernel and counter. □

**Example 4.2** (The first counter that must be retained)**.** For $\lambda = \omega^3$, formula (4.1) gives $J_\lambda(t) = X^3_0(t) + E^3_0(t)$. For $\lambda = 2\omega^3$, it gives $J_\lambda(t) = J_{\omega^3}(X^3_4(t)) + E^3_4(t)$, since $N(\omega^3) = 4$. Replacing either expression by its outer $X$ alone would change exact graph equations and could change the result of cancellation.

## 4.1 Separation with indices allowed to grow

We need a uniform estimate, because the block recursion creates retained masses comparable with $f(t)$. Merely proving a tail for each fixed mass would not license the induction.

**Lemma 4.3** (Uniform kernel separation)**.** *Put $C_d = 1000d^2$, $T_3 = 16$, and for $d \geq 4$ put $T_d = \max(T_{d-1}, 100C_{d-1} + 100d)$. For $d \geq 3$, $c, e \geq 0$, signed $j, k$, and $R \geq 0$, suppose*

$$t \geq T_d, \qquad f(t) \geq C_d(c + e + |j| + |k| + R + 1). \tag{4.5}$$

*Then the depths are defined and at least $f(t)$, and*

$$\begin{aligned} &X^d_{c,j}(t+1) > f^R(X^d_{e,k}(t)), \\ &\quad c < e \quad \Longrightarrow \quad X^d_{c,j}(t) > f^R(X^d_{e,k}(t)). \end{aligned}$$

*At equal input and equal $c$, exact order is order of $j$. On a common computed tail, equality of two rank-$d$ values occurs exactly when input, $c$, and $j$ are identical.*

*Proof.* We first record the elementary degree-two estimate. If $A = f(w)/2 \geq |j| + |k| + R + 4$, then

$$V_j(w+1) > f^R(V_k(w)).$$

Indeed $f(w+1)/2 - A \geq A$, so the left iterate starts at a larger seed and has more iterations than the right one plus $R$. At equal input, the $V$ values are ordered exactly by their indices. This argument permits indices and required depth varying with a later induction parameter, provided the displayed numerical bound is paid.

For $d = 3$, $B = f(t)$ is even and $B \equiv 1 \pmod 3$; write $a = (B-1)/3$, which is odd. The three initial entries for prefix classes $\delta = 0, 1, 2$ are

| $\delta$ | initial $c$ | entry norm | spent count |
|---|---|---|---|
| 0 | $B - 1 = 3a$ | $B$ | 1 |
| 1 | $B = 3a + 1$ | $B$ | 1 |
| 2 | $B - 2 = 3(a-1) + 2$ | $f(B)$ | 2. |

Descending $c$ by three applies $V_{-\lceil (c-3)/2 \rceil}$ to the preceding entry and adds its displayed iteration length to the counter. Write $Y^\delta_r = X^3_{3r+\delta}(t)$. At the first common entry $r = a - 1$, $Y^0_r = f(Y^1_r)$,

since the two subtracted ceilings differ by one. One further aligned block amplifies this integer input gap to any fixed $f$-depth by the degree-two estimate. Subsequent indices are favourable: the class-zero index is no smaller than the class-one index. Next, $Y^2_{a-1} = f(B)$ whereas

$$Y^1_{a-1} = f^\ell(B), \qquad \ell = f(B)/2 - \lceil (3(a-1)+1)/2 \rceil + 1.$$

For $a$ larger than the desired depth by ten, $\ell$ exceeds that depth plus one. Finally align class two at entry $r-1$ with class zero at entry $r$. The initial seeds satisfy $f(B) > B$, and $\lceil (3r-4)/2 \rceil \leq \lceil (3r-3)/2 \rceil$, so the degree-two estimate again applies in the favourable direction. Consequently

$$Y^0_r > f^L(Y^1_r), \quad Y^1_r > f^L(Y^2_r), \quad Y^2_r > f^L(Y^0_{r+1})$$

at all required target entries, after paying the fixed depth $L$. All smaller block entries are at least $B$, and the absolute degree-two indices are at most $\lceil 3a/2 \rceil + 2$ plus fixed alignment shifts; $f(B)/2$ pays their sum and $L$. Condition (4.5), with $C_3 = 9000$, leaves more than ten blocks beyond the target and the required $R + |j| + |k| + 1$ depth. This proves the same-input statement by chaining the three adjacent classes.

For cross-input cubic comparison, $B' = f(t+1) > 2^B$ supplies more than one extra leading block. At the smaller process's starting coefficient, the larger process has completed a quadratic block and has seed at least $f(B') > f(B)$, exceeding either possible smaller seed. Align the remaining blocks, shifting the coefficient alignment by at most $\max(\lfloor c/3 \rfloor - \lfloor e/3 \rfloor, 0)$. Their indices are bounded by $\lceil 3a/2 \rceil + 2$ plus this shift, and their differences are bounded by twice the shift plus three. The same degree-two estimate at smaller entry at least $B$ pays each comparison. Extra remaining blocks only increase the final norm. This proves the cubic cross-input statement.

For the induction $d \geq 4$ set $L = R + |j| + |k| + 1$, $B = f(t)$. Condition (4.5) leaves at least $B/(2d) > L + 2$ leading blocks. Every aligned lower-rank mass is at most $2B$, and both its pair sum and each self-pair sum, together with $L+1$, are at most $5B$. All smaller aligned entry norms are at least $B$. Since $B \geq T_d \geq 100 C_{d-1}$,

$$f(B) \geq 2^B > 5 C_{d-1} B.$$

Thus the induction hypothesis is valid at the *smaller* entry norm even though the new masses may be comparable with $B$.

At equal $t$ and $c < e$, align the two descents at the initial leading coefficient of the $e$ process. If no extra leading block was needed, the $c$ process's remainder is at least as large, with smaller retained mass. If extra blocks were needed, its last extra full block exceeds the other initial remainder. These facts order their aligned seeds weakly. To justify this seed comparison explicitly, maximal predecessors are monotone in both ordinal bound and norm budget. If entry norms $v \geq w$ and masses $a, b$ satisfy $f^i(v) - a \geq f^i(w) - b$ at every step, the first process stays above the second at simultaneous steps, so takes at least as many steps and has at least as large a terminal norm. Decreasing mass at equal entry is therefore favourable. A larger entry also pays an adverse mass difference whenever $f(v) - f(w) \geq a - b$, since successive iterate gaps do not decrease. At aligned block $k$ the lower masses are $c + d(k-1) < e + d(k-1)$. Equal seeds use lower-rank same-input separation; distinct seeds use lower-rank cross-input separation. The estimates just proved pay both, including lower self-pairs. One such block gives depth $L$, and the rest preserve it.

At $t+1$, its first budget $B' > 2^B$ gives an extra full leading block before alignment. Its entry is at least $B'$ and retained mass at most $B'$. The smaller initial remainder is below that block,

has entry $B$, and retained mass at most $B$. The inequality $f(B') - f(B) > B'$ pays even an adverse retained mass; hence the larger seed is strict. At subsequent aligned blocks the mass pair is bounded by $4B$; lower-rank cross-input separation, including its self-pair bounds, applies as before. This proves the cross-input statement. Monotonicity at fixed mass extends adjacent comparison to arbitrary larger inputs.

Finally the number of remaining blocks exceeds $|j| + |k| + 1$, so signed depths leave a positive iterate and output at least $f(t)$. Taking the maximum of the cross-pair and the two self-pair thresholds excludes every unequal-input or unequal-mass equality. The coherent identity settles the remaining equal-input, equal-mass case. □

### 4.2 Counter leaders, exact ties, and residues

For $r \geq 3$ define $A_r = t$ and successively

$$A_{h-1} = X_{c+h}^{h}(A_h) \quad (h = r, r-1, \ldots, 4), \qquad u = X_{c+3}^{3}(A_3), \qquad H = f(u).$$

We call $H$ the counter's *leader*; $u$ is its leader preimage.

**Lemma 4.4** (Counter expansion and collision)**.** *On an effective tail,*

$$E_c^r(t) = \frac{H}{2} + 1 - \lceil c/2 \rceil + \sum_{h=3}^{r} E_{c+h}^{h}(A_h), \qquad \frac{H}{2} < E_c^r(t) < \frac{3H}{4}. \tag{4.6}$$

*Suppose counters of ranks $r \geq s \geq 3$ have equal leaders. Then they have the same prefix index $c$. If $r = s$ their whole values are equal; if $r > s$ the smaller-rank input is the corresponding $A_s$, and*

$$E_c^r(t) - E_c^s(A_s) = \sum_{h=s+1}^{r} E_{c+h}^{h}(A_h). \tag{4.7}$$

*Every counter introduced on the right has rank strictly greater than $s$. For distinct leaders, counter order agrees with leader order. Equal leaders are ordered by counter rank, with equality at equal rank.*

*Proof.* Iterate (4.4) until the cubic term; its bottom quadratic count is $f(X_{c+3}^{3}(A_3))/2 + 1 - \lceil c/2 \rceil$. This proves the exact identity. A descent with initial norm $w$ and $L$ steps satisfies $L \leq f^L(w) - w$, since each iterate increases its integer argument. Thus each residual above rank three is less than its next entry norm, hence less than $u$. The cubic residual plus the fixed correction is also less than $u$, since it is at most $u - A_3 + 1 - \lceil c/2 \rceil$. It is positive on a computed tail: the number of cubic blocks tends to infinity. Therefore

$$H/2 < E_c^r(t) < H/2 + (r-2)u < 3H/4$$

once $f(u) > 4(r-2)u$ and the finite correction bound is paid.

Equality of leaders first cancels $f$, then compares the two outer cubic kernels $X_{c+3}^{3}$. Lemma 4.3 forces equality of their prefix indices and arguments. Cancel the equal rank-four kernels next, then rank five, up to rank $s$. If $r = s$ the original inputs are equal. If $r > s$ the rank-$s$ input is precisely $A_s$. Subtraction of the exact recursions gives (4.7). Its right side is positive and tends to infinity. For different leaders $H_1 < H_2$, their being distinct values of $f$ gives $H_2 \geq 2^{H_1+1}$; the half-core bounds make the order strict. This also proves the assertions with arbitrary fixed outer shifts after raising the computed tail. □

**Lemma 4.5** (Effective residues of kernels and counters)**.** *For each fixed $d, c, j, m$, both $X^d_{c,j}$ and $E^d_c$ have an effectively computable constant residue modulo $m$ on a computable tail. The $X$ residue is $r_m$. A valid counter residue is*

$$E^d_c(t) \equiv p_d\big((r_{Q_d} - c) \bmod Q_d\big) \pmod m, \qquad Q_d = d!m^{d-1}, \tag{4.8}$$

*where $p_d(n)$ counts partitions of $n$ into parts at most $d$.*

*Proof.* A coherent $X$ lies in the range of $f$ with preimage at least $t$ on a computed tail. Apply Lemma 2.2. For the counter, the restricted-partition and congruent-budget theorems in [7] prove the following two facts, with full degree inductions: $Q_d$ is a period of $p_d(n)$ modulo $m$, and a controlled $\omega^d$ block with retained norm $c$, whose successive budgets are all congruent to $S$ modulo $Q_d$, has complete count congruent to $p_d(S - c)$ modulo $m$. Budgets may grow; equality of the budgets is not assumed. For $t \geq Q_d$, every actual budget $f^i(t)$ is congruent to $r_{Q_d}$, by Lemma 2.2. Choose $S \geq c$ with that residue and apply the theorem. The partition period reduces its argument to the representative in (4.8). Finite dynamic programming computes $p_d(n)$ for all $n < Q_d$. The domain bound and $t \geq Q_d$ provide an explicit threshold. This proof of residues uses neither comparison nor an existential decision procedure. □

## 5 A closed existential decision procedure

Fix a finite maximum degree $D \geq 3$. A *word* in one raw variable is a finite composition of signed constant shifts and the functions

$$f, \quad V_j, \quad X^r_{c,j}, \quad E^r_c \quad (3 \leq r \leq D),$$

with fixed integer indices satisfying the preceding conventions. A word-linear row is an integer linear combination of finitely many such words, evaluated in possibly different raw variables, plus a constant. We allow arbitrary Boolean combinations of row comparisons with zero and fixed-modulus congruences. Each raw variable has an explicit individual lower bound ensuring that all its words are defined.

Every word has a computable domain tail and a lower offset $M$ with $W(n) \geq n - M$. This follows structurally: shifts add to the offset; $f$, $V$, and coherent $X$ dominate their input on their stated tails; counters do so as well. For the latter, a sufficiently large degree-three block contains two quadratic blocks, and a quadratic block entered at least at $f(t)$ already spends more than $f(t)$ steps. Higher-degree blocks contain one of these. These structural bounds also pay every threshold at an interior argument of a composition. They are not obtained by treating the interior argument as a raw variable.

**Lemma 5.1** (Ordered maximum-variable guards)**.** *For two words $A(x), B(y)$ and the guard $x \geq y$, an effective common tail permits their three-way comparison to be rewritten as a finite Boolean combination in which $x$ occurs only linearly and words are applied only to the remaining variable $y$. For two words at the* same *raw variable the corresponding tail comparison is effectively decided. All thresholds and newly generated words are computed before the tail is used.*

*Proof.* Peel fixed outer shifts. Unequal range values have gaps eventually larger than a fixed shift. Lemma 4.4 gives the same fact for two counter values, including equal-leader unequal-rank values

whose exact residual difference tends to infinity. For a counter with preimage $u$ and range value $f(v)$, its half-core bounds give

$$E_c^r < f(v) \iff u \le v, \qquad E_c^r > f(v) \iff u > v;$$

the gaps again tend to infinity. Thus fixed shifts are kept for exact-core ties and otherwise cannot close the gap.

For $f/f$ strip the two $f$ nodes. For $f$ against a coherent range kernel, move one fixed depth to its index and strip $f$; for example $V_j = f \circ V_{j-1}$. Two $V$ heads compare lexicographically by argument, then by index, using the elementary estimate in the proof of Lemma 4.3. Two equal-rank $X$ heads compare by argument, then by $(-c, j)$, using that lemma and retaining the exact equal-input branch.

For $X^r$ against $X^s$ with $r > s \ge 3$, expand only the higher-rank head by (4.3) until its outer rank is $s$, and immediately compare and strip that head against the opposite rank-$s$ head. If $s = 2$, the exposed quadratic head is $V_{j-\lceil c/2 \rceil}$ and the same procedure uses its index comparison. No inverse counter is introduced. For counter/range comparison use the displayed preimage rule. For counter/counter comparison use their leader preimages followed by the rank and exact-tie rule of Lemma 4.4.

Here is a termination measure, needed because mixed-rank expansion can increase the written expression. First count all $E$ nodes in the two words. Replacing a counter by its leader-preimage composition removes its outer $E$ node and retains its argument once. For the range part, assign each rank-$s$ occurrence weight $\omega^{D-s}$, where $f$ has rank one and $V$ rank two, and naturally sum these weights. A mixed comparison deletes the opposite rank-$s$ occurrence and introduces only finitely many occurrences of ranks strictly above $s$, retaining the original higher-rank occurrence. Its ordinal weight decreases. Equal-rank stripping and $f$ cancellation decrease weight as well. The lexicographic pair (number of $E$ nodes, range weight) is therefore well-founded along every recursive child. There are finitely many branches at each step.

Ultimately the $x$ side is a shifted leaf, so its comparison with a word at $y$ is linear in $x$; or the $y$ side is a shifted leaf and the $x$ side is nonlinear. In the latter case the nonlinear value is at least $2^{x-M}$ for a computed $M$, so eventually exceeds $y$ plus any fixed shift, since $y \le x$. When both words have the same raw variable, continue stripping until leaves or leaf/nonlinear growth decides the comparison; do not create a purported linear atom $x < W(x)$. Take the finite maximum of all local bounds using the structural offsets of interior arguments. The finite recursion tree, rather than a guessed fixed envelope of indices, computes that maximum. □

The guards compare individual word values. A linear row can still cancel its largest values, so it requires a different induction.

**Lemma 5.2** (Ranked cancellation)**.** *For a finite collection of word-linear rows and a chosen maximal raw variable $x$, one can compute a tail and finite guarded cases in which each row either has a known strict sign or is exactly linear in $x$. The guards can themselves be made linear in $x$ by Lemma 5.1. No correlation between word values is discarded.*

*Proof.* Absorb fixed outer shifts into the constant and keep raw-linear terms separate. The other top-level atoms are ordinary range values (rank zero for this row argument) and counters of ranks $3, \dots, D$. Partition their ordinary values and counter leaders into ordered equality classes, retaining every equality and order guard. Merge equal-rank counters in a class as *whole counters*,

justified by Lemma 4.4. Prefer a representative actually involving $x$ when one remains in the current class. A class now has ordinary coefficient $a$ and at most one coefficient $b_r$ for each counter rank. Its leading coefficient is

$$a + \tfrac{1}{2} \sum_r b_r.$$

If it vanishes, pay its residuals before deciding any sign. When $a \neq 0$, expand the counters by (4.6); the common half-leader cancels the ordinary atom, leaving only residual counters and constants. When $a = 0$, let $s$ be the least surviving counter rank. Rewrite each higher counter by (4.7) as the common rank-$s$ counter plus higher-rank residuals. Since $\sum b_r = 0$, the common term cancels. Remove an entirely zero class. Repartition after every rewrite, since its newly exposed leaders may coincide with leaders elsewhere in the row.

Count *top-level* row atoms, assigning the ordinary rank-zero atom weight $\omega^D$ and a rank-$r$ counter weight $\omega^{D-r}$. Natural sum is a well-founded measure. The first rewrite loses a rank-zero atom and adds only positive-rank atoms. The second loses the least rank $s$ and adds only ranks strictly greater than $s$. Existing higher-rank atoms can proliferate, but cannot offset that ordinal decrease. Nested nodes inside their arguments are not included in this measure; their comparisons terminate by the separate guard measure. Every branching row rewrite therefore strictly decreases weight. The tree is finitely branching and has no infinite branch, so it is finite. Enumerating it to completion computes all generated words, coefficients, prefix indices, and guard thresholds before any finite-prefix case is split off.

After the rewrites every surviving leading coefficient has absolute value at least $1/2$. Let $P = f(u)$ be the greatest surviving leader, $S$ the current absolute nonlinear coefficient mass, and $A, B$ bounds for the absolute raw-linear coefficients and constant. The exact counter corrections have total size at most $DSu$. Every smaller leader $Q$ satisfies $Q < \log_2 P$, so its total contribution is at most $S \log_2 P$. For $u \geq 8DS + 16$ each of these two errors is less than $P/16$.

If a current nonlinear $x$ occurrence survives, the greatest leader is at least its leader. Structural offsets therefore give $u \geq x - M$ for a computed $M$. Require $x \geq 2M + 16$ and $x \geq 8(A + B + 1)$. The elementary inequality $2^{x-M} \geq x^2$ on this range, proved at the initial value and then by doubling, gives $Ax + B < P/8$. Total error is less than $P/4$, while the leading magnitude is at least $P/2$. Hence its coefficient strictly determines the sign. If no current nonlinear $x$ occurrence remains, keep the exact row, which is linear in $x$.

Growth provenance refers only to current occurrences. A deleted leader passes no $x$ label to a residual which does not actually contain $x$. This prevents an invalid sign decision after cancellation has removed all nonlinear dependence on $x$. Take maxima of these explicit bounds over the finite row and guard trees. Ordered partitions retain the original shared arguments throughout, even when an inconsistent branch is later discarded. □

**Theorem 5.3** (Closed existential word arithmetic)**.** *The existential closure of every Boolean combination of word-linear row comparisons and fixed-modulus congruences in the finite-degree language above is decidable, on its explicit word domains.*

*Proof.* First remove a variable which occurs only linearly, if any, by ordinary Presburger quantifier elimination. Words in other variables are parameters in this step; the eliminated variable occurs in none of them. The output is still a Boolean combination of the same kind. Repeat or evaluate the closed matrix if no variables remain.

Otherwise branch on a maximal raw variable $x$, retaining $x \geq x_i$. Compute the complete finite guard and row trees of Lemmas 5.1–5.2. Compute residue thresholds for each congruence as well: Lemma 4.5, composed with interior lower-offset bounds, makes every nonlinear word's residue constant. Collect shifts in its fixed residue. On that tail, congruences are thus congruences in raw variables only. Take a common $N$ paying all original and generated domains, sign estimates, guards, and residue bounds.

Before executing any tail transformation, split the *original current matrix* into the all-raw-inputs-$\geq N$ branch and finitely many branches fixing some raw $x_i = c < N$ above its original lower bound. In a finite branch substitute $c$ into its original words and evaluate them exactly. Do not evaluate a newly introduced coherent negative-depth word below the tail on which it was introduced. Exact kernel evaluation is finite by the block recursion. Each such branch removes a quantified variable.

On the all-large branch, use the guarded row and congruence transformations. Everything now contains $x$ only linearly. Presburger elimination of $\exists x$, treating words in the remaining variables as parameters, again produces word-linear rows and congruences with one fewer variable. Preserve all Boolean connectives and every equality/order guard. No word has been applied to a sum, and no auxiliary derived value has been made an independent witness.

The induction measure for the whole algorithm is the number of raw existential variables. It strictly decreases in every finite-prefix, pure-linear, and all-large branch. The subsidiary guard and row trees are finite by their own measures. Consequently this is a terminating effective procedure. No feasible complexity estimate follows: finite exception values and arithmetic thresholds can be extremely large. □

This theorem is deliberately about *closed existential* sentences. In a formula with a free parameter, that parameter could be the largest raw argument and could not be removed by the stated induction. We have proved neither free-parameter quantifier elimination nor the full first-order theory of arbitrary polynomial-collapse functions.

## 6 Exact polynomial graphs and the internal-level-two consumer

**Theorem 6.1** (Polynomial collapse graphs)**.** *For each fixed nonzero polynomial limit $\lambda < \omega^\omega$, the graph $y = h_\lambda(n)$ is effectively expressible as a finite disjunction of word-linear equations in $n, y$, with explicit domains and exact finite exceptions. One finite degree bound suffices for each finite collection of such graphs. Arbitrary fixed nesting and addition of these functions admit a total functional existential graph translation.*

*Proof.* Remove one unit at the least positive degree of $\lambda$, writing $\lambda = \mu + \omega^d$, and put $c = N(\mu)$. There are finitely many monomial units. Starting with $2J_0(t) = 2t$, compile (4.1) recursively: substitute the unary word $X_c^d(t)$ for the input in the already compiled expression for $2J_\mu$, and add $2E_c^d(t)$. For $d = 1$ use $X = f$, $2E = 2$; for $d = 2$ use

$$X = V_{-\lceil c/2 \rceil}, \qquad 2E = f(t) - 2\lceil c/2 \rceil + 2;$$

for $d \geq 3$ retain the $X$ and $E$ heads. This substitution composes unary words only. It never applies a function to a sum of compiled values.

Finally replace $t$ by $n + N(\lambda)$. On a computed tail this yields $2y = P_\lambda(n)$ with an integer word-linear $P_\lambda$. The finitely many excluded $n \geq 0$ are evaluated by the total closed predecessor

algorithm and included as disjuncts $n = k$, $y = h_\lambda(k)$. The graph is therefore total and functional on the original natural domain. To flatten a nested expression, introduce a fresh natural variable for each arithmetic node: addition and fixed multiples have linear equations; a collapse node has the graph just constructed, with its original argument variable. Preserve all repeated original variable names and quantify every new node variable. This creates no independence assumption: the graph equations enforce precisely the original correlations. □

**Theorem 6.2** (Canonical source comparison through level two)**.** *Universal weak, WW-strict, and equality comparison is decidable for original canonical $T$ assignments with ground free variables and maximum internal type level at most two. The maximum includes every subterm and binder type, not merely the result and free-variable types.*

*Proof.* The actual-source induction in [3] gives the forward residual translation. Its level bound can be seen directly in that generated grammar. Coordinate two has no exponent constructor: one would require a coordinate three. Its closed values are therefore natural linear combinations of 1 and $\omega$. Coordinate one forms natural sums and products and powers $2^{a\omega+b} = \omega^a 2^b$ for finite $a, b$; its values are polynomials in $\omega$. Each direct scalar head $\omega \otimes c$ is consequently a polynomial limit below $\omega^\omega$. Temporary functional binder variables disappear under the canonical abstraction substitution, and the internal-level hypothesis includes their types. No statement about arbitrary ambient coordinate classes is needed.

Evaluate and compare closed positive coordinates. Translate a zeroth counterexample to the requested universal relation using Theorem 6.1; all fresh graph variables are quantified, with nonnegative domains, and original variables remain positive. The result is a closed existential Boolean combination in a finite-degree word language. Theorem 5.3 decides whether such a counterexample exists. This separately decides each of the source's strict, weak, and equality relations with its actual coordinate convention. There is no fixed degree bound across all input terms: the finite degree needed is computed from each pair. □

## 7 Arbitrary fixed heads: sparse endpoints

Polynomial graphs were needed for completely unrestricted arrangements of polynomial heads. Different arrangements permit arbitrary fixed heads without constructing their graphs. This section isolates the spacing fact used by all of them.

**Lemma 7.1** (Sparse endpoint comparison)**.** *For a fixed nonzero limit $\lambda$, $J_\lambda$ is strictly increasing. Its distinct norm inputs have distinct terminal endpoints, ordered in the same way. For any two distinct endpoints $Z < W$ one has*

$$W \geq 2^{Z+1}. \tag{7.1}$$

*Every value with endpoint $Z$ lies strictly between $Z$ and $2Z$ and below the next value of $f$.*

*Proof.* Strict increase is the controlled finite-tail strictness proved in [3]: when $t < u$, the smaller ordinal has norm $t \leq f(u)$, so the controlled comparison premise holds. Write $J_\lambda(t) = f^L(t) + L$ and $J_\lambda(u) = f^M(u) + M$. If their endpoints were equal, injectivity of $f$ would force $L > M$; otherwise cancelling $L$ iterates would give $t \geq u$. The equal endpoints and $L > M$ would imply $J_\lambda(t) > J_\lambda(u)$, a contradiction. Put $s = f^{L-1}(t)$. Since $t \geq N(\lambda) \geq 2$, one has $L \leq s - 1$, so $f(s) < J_\lambda(t) < 2f(s) < f(s+1)$ by Lemma 2.1. These intervals are disjoint and preserve order. If $W > Z = f(s)$ is another endpoint, then $W \geq f(s+1) \geq 2^{Z+1}$. □

### 7.1 Nested common-head sums

Fix any finite list of fixed nonzero limits $\lambda_1, \ldots, \lambda_l$. Let $\mathcal{A}_0$ consist of natural constants, positive raw variables, addition, and $f$, with arbitrary fixed nesting. Recursively set

$$\mathcal{A}_r = \left\{ c + \sum_i a_i h_{\lambda_r}(B_i) : c \geq 0,\ a_i > 0,\ B_i \in \mathcal{A}_{r-1} \right\}. \tag{7.2}$$

The sum can be empty. Each input supplies this finite level certificate. There is no open remainder outside the head sum at a positive level. For example $x + h_\lambda(y)$ does not pass that positive-level certificate merely because both $x, y$ are allowed at level zero. Heads at different levels may be completely different, and may exceed $\omega^\omega$.

**Theorem 7.2** (Common-head layers)**.** *For $E, F \in \mathcal{A}_r$, each pointwise relation $E < F$, $E = F$, $E > F$ has an effectively constructed formula in $(\mathbb{N}; 0, 1, +, <, f)$ with the same raw free variables and exactly the same truth at every positive valuation. Hence arbitrary Boolean and quantified combinations of certified comparisons are decidable.*

*Proof.* Subtract the two expressions and write the row as

$$Q = K + \sum_{i<m} a_i J_{\lambda_r}(t_i), \qquad t_i = N(\lambda_r) + B_i.$$

Coefficients and $K$ are signed integers. With no remaining atoms the sign is the computable sign of $K$. Otherwise set $C = \sum_i |a_i|$ and $L = \max(4, 2C, 4|K| + 4)$. On the branch with all $t_i \geq L$, partition the inputs by equal values, combine coefficients, and delete zero blocks. If every block cancels, $Q = K$. Otherwise its sign is the coefficient of the greatest surviving input. Indeed let $W$ be its endpoint and $Z$ the greatest lower endpoint, if any. Lemma 7.1 gives $W \geq 2^{Z+1} \geq 2Z^2 \geq 4CZ$, since $Z \geq L \geq \max(4, 2C)$. All lower contributions have total absolute value less than $2CZ \leq W/2$. Also $|K| < W/4$, whereas the leading contribution has absolute value greater than $W$. Its sign cannot change. Larger canceled blocks play no part in this estimate. Enumerating the finitely many total preorders of the labeled inputs produces a finite formula, guarded by $t_i = t_j$, $t_i < t_j$, and $t_i \geq L$.

For the other branch choose a small remaining index, enumerate $N(\lambda_r) \leq c < L$, retain its exact guard $t_i = c$, evaluate the closed $J_{\lambda_r}(c)$, fold $a_i J_{\lambda_r}(c)$ into $K$, and remove that occurrence. Recompute $L$ and repeat. Each small call removes one occurrence; its possibly enormous updated constant is still a fixed computable integer. Overlapping branches are harmless in a disjunction. No open value is folded into a constant before its whole argument is fixed.

Induct now on $r$. At level zero comparisons are base-language atoms. At higher levels every guard compares preceding-level expressions with each other or with constants; a proposed constant below $N(\lambda_r)$ gives a constant truth value. Thus every generated predicate is covered by the induction hypothesis. The lexicographic pair (level, remaining head occurrences) proves termination: small folding decreases the second, guard translation the first. Shared variables remain shared throughout. The result is an exact base formula, to which Theorem 2.3 applies after any desired raw quantifiers are added. □

**Example 7.3** (Cancellation changes the controlling input)**.** Consider $2h_\lambda(x) + h_\lambda(y) - 3h_\lambda(z)$. On the computed large-input branch, a unique largest argument determines the sign by its coefficient. But on the guard $x = z$, the row becomes exactly $h_\lambda(y) - h_\lambda(x)$, whose sign is that of $y - x$;

when all three arguments agree the row is zero. The algorithm retains those guards rather than treating three head values as independent large numbers. Small inputs are evaluated by the finite folding procedure, so this is an exact comparison theorem rather than only an eventual rule.

### 7.2 Different heads in one control interval

**Lemma 7.4** (Head order in a strict interval)**.** *Let $\lambda > \mu$ be nonzero fixed limits. If $t \geq \max(N(\lambda), N(\mu)+2)$ and $N(\mu) \leq u < f(t)$, then $J_\lambda(t) > J_\mu(u)$ and their endpoints differ. Consequently, if finitely many norm inputs $t_i$ have minimum $s \geq \max_i N(\lambda_i)+2$ and $\max_i t_i < f(s)$, their outputs are ordered lexicographically by (ordinal head, norm input). Equality requires equality of both, and different output blocks have distinct sparse endpoints.*

*Proof.* The least limit above $\mu$ is $\mu+\omega$, of norm $N(\mu)+2$. Controlled strictness at common norm $t$, or equality of the two heads, gives

$$J_\lambda(t) \geq J_{\mu+\omega}(t) = J_\mu(f(t)) + 1 > J_\mu(u).$$

The exact middle identity follows by removing the final $\omega$ block: its first predecessor retains $\mu$, assigns the remaining full budget to its finite tail, and has total norm $f(t)$. If the two displayed outputs had equal endpoints, their complete counts would satisfy $L > M$; canceling the common $f$ iterates would give $u = f^{L-M}(t) \geq f(t)$, a contradiction. For the finite collection, the lower input is below $f(s) \leq f(t)$ for every higher-head input $t$. Apply this argument to different heads and Lemma 7.1 within the same head. □

The interval inequality is strict for a reason. At $u = f(t)$,

$$J_{2\omega}(t) = f^2(t) + 2, \qquad J_\omega(u) = f^2(t) + 1.$$

Their endpoints coincide, and $J_{2\omega}(t) - 2J_\omega(u) = -f^2(t) < 0$ despite the positive coefficient of the greatest ordinal head. An algorithm using head order at that boundary would therefore be incorrect.

## 8 Balanced heterogeneous layers

The next certificate permits different sibling heads and an open lower-level remainder. At level zero use the same $\mathcal{A}_0$ as before. At a positive level choose a finite common preceding-level basis $B_1, \ldots, B_k$. It may have repeated or dependent entries. New expressions are nonnegative affine combinations of the $B_j$ and atoms

$$h_\lambda\left(c + \sum_j a_j B_j\right), \qquad c \geq 0, \quad a_j \geq 1. \tag{8.1}$$

Each atom may have its own arbitrary fixed head and coefficients. The essential condition is that *every head argument includes every common basis entry positively.* The basis is checked on the whole pair being compared, including its outside remainder. A variable appearing only in that remainder is not admitted by ignoring it while choosing the basis. Zero multiples are discarded when computing actual supports.

**Theorem 8.1** (Balanced comparison)**.** *For two expressions with one finite balanced certificate, their three-way pointwise comparison has an effectively constructed base formula in* $(\mathbb{N}; 0, 1, +, <, f)$ *with exactly the same free variables. Arbitrary Boolean and quantified combinations of such comparisons are decidable.*

*Proof.* A signed comparison row has the form

$$Q = K + \sum_j r_j B_j + \sum_{i<m} q_i J_{\lambda_i}(t_i), \qquad t_i = b_i + \sum_j a_{ij} B_j, \quad b_i = N(\lambda_i) + c_i. \tag{8.2}$$

The head-free case is already a signed affine comparison at the preceding level. Otherwise put

$$S = \sum_j B_j, \quad C = \sum_i |q_i|, \quad C_0 = \max_j |r_j|, \quad A = \max_{i,j} a_{ij}, \quad B = \max_i b_i,$$

with $C_0 = 0$ for an absent remainder and $A = 1$ for an empty basis. Set

$$L = \max\{16, 2C, 4C_0 + 4|K| + 4, A + B + 2, \max_i N(\lambda_i) + 2\}. \tag{8.3}$$

On the branch $s = \min_i t_i \geq L$ one has

$$S \leq t_i \leq AS + B, \qquad \max_i t_i \leq As + B < s^2 \leq 2^s \leq f(s).$$

Thus Lemma 7.4 orders all head outputs by head and then input. Partition the inputs within each equal head, combine coefficients, and discard zero blocks. If all cancel, the remaining row is exactly the preceding-level $K + \sum r_j B_j$.

If a block survives, let $W$ be the endpoint of the last one and $v$ its norm input. Its coefficient has absolute value at least one, so its contribution has magnitude greater than $W$. The sparse-endpoint estimate bounds all lower head contributions by $W/2$, exactly as in the proof of Theorem 7.2. The outside remainder is bounded using this *surviving* input, not a possibly canceled maximum:

$$\left| K + \sum_j r_j B_j \right| \leq |K| + C_0 v < v^2/4 \leq f(v)/4 \leq W/4.$$

The surviving last coefficient therefore determines the strict sign. The finitely many guarded preorders within equal heads express this rule by comparisons of affine preceding-level expressions.

We must retain every small exception. Every positive expression in the certificate satisfies $B(\vec{x}) \geq x$ for each raw variable in its actual support. Induct through addition and $f$, and at a head use $h_\lambda(e) > e$: the finite ordinal $e$ is below $\lambda + e$ and has norm $e \leq f(N(\lambda) + e)$, so controlled strictness and $\psi(e) = e$ apply. Fixed positive multiples preserve the property. If some $t_i < L$, positivity of every $a_{ij}$ implies $S < L$. Therefore every raw variable anywhere in the basis, including the outside remainder, is less than $L$. Enumerate the finite box $[1, L-1]^V$ on its *actual* support $V$; evaluate each closed row and its inputs exactly, retaining precisely points where some $t_i < L$ and the requested sign holds. Variables outside $V$ remain unrestricted; empty support gives a single empty valuation.

All large-branch guards, and the all-canceled remainder, are affine comparisons of the $B_j$ and thus have their preceding-level certificate. The finite-point exception formula is already in the base language. Induction on certificate level proves the claimed translation and termination. Theorem 2.3 supplies its arithmetic decision. Repeated basis entries and shared variables never become independent parameters in this construction. □

**Example 8.2** (A lower-head argument may be larger)**.** Let $E, F$ have a common preceding-level certificate and $\lambda > \mu$. The row

$$3h_\lambda(E+F+2) - 2h_\mu(2E+F+1) - F$$

is balanced on basis $(E, F)$. Its two norm arguments need not have the same order as their ordinal heads, and $F$ remains open outside them. Nevertheless, once both norms exceed (8.3), the row is positive by the higher-head coefficient. Its exact comparison is completed by the finite support box. Omitting $F$ from one head argument would remove the inference that a small input bounds the entire open remainder.

# 9 One arbitrary core: direct Presburger comparison

An especially useful specialization no longer requires a certificate for the interior core. Let $E$ be any expression made from natural constants, positive raw variables, addition, fixed nonnegative multiples, and arbitrary fixed nonzero-limit functions $h_\lambda$, with arbitrary finite nesting. No subtraction or exposed $f$ constructor occurs inside this stated grammar. The only property needed below is $E \geq x$ for every variable in its actual support, proved by the positive-source induction just given. It need not have a balanced-level certificate, nor range independently over all naturals.

**Theorem 9.1** (Arbitrary single-core rows)**.** *For the signed row*

$$Q(\vec{x}) = K + rE(\vec{x}) + \sum_i q_i h_{\lambda_i}(a_i E(\vec{x}) + c_i), \quad a_i \geq 1,\ c_i \geq 0, \tag{9.1}$$

*with fixed integer $K, r, q_i$ and fixed nonzero limit heads, each sign predicate $Q < 0$, $Q = 0$, $Q > 0$ has an effectively computable finite Presburger formula in the original raw variables. Thus universal comparison of these rows needs no arithmetic decision theorem about $f$.*

*Proof.* Collect identical triples $(\lambda_i, a_i, c_i)$ and discard their zero coefficients. If any heads remain, use (8.3) with the single formal basis $E$, $C_0 = |r|$, and $b_i = N(\lambda_i) + c_i$. Increase that bound to

$$T = \max\bigl(L, 1 + \max_{i,j} |b_i - b_j|\bigr).$$

For $E \geq T$, the norms lie in one strict $f$ interval. Within equal heads their affine inputs are ordered lexicographically by $(a_i, b_i)$: a difference of integer slopes dominates the fixed offset difference. Equal triples have already been collected. Lemma 7.4 and the remainder estimate in the balanced proof therefore show that the sign is the coefficient of the greatest triple $(\lambda_i, a_i, b_i)$. This use of the balanced *numerical estimate* requires no certificate or comparison algorithm for $E$ itself. If no heads remain, use $T = \max(1, |K| + 1)$; the final sign of $rE + K$ is $\operatorname{sign}(r)$ when $r \neq 0$, otherwise $\operatorname{sign}(K)$. Denote the computed final sign by $\sigma$.

Let $V$ be the actual raw support of $E$. Empty support is handled by one closed evaluation. Otherwise, outside $[1, T-1]^V$ there is an $x \geq T$, and positivity gives $E \geq T$. Evaluate every row in the finite box exactly and record its sign. For a desired sign $s$ the exact formula is

$$\left(\bigvee_{x \in V} x \geq T\right) \wedge [\sigma = s] \quad \vee \bigvee_{\substack{v \in [1,T-1]^V \\ \operatorname{sign} Q(v) = s}} \bigwedge_{x \in V} x = v(x).$$

It is a finite Presburger formula on the original variables. The box is effective and each closed evaluation terminates. If the tail falsifies a universal comparison, a concrete admissible counter-valuation sets one supported variable to $T$ and all others to one; the dominance property certifies it without evaluating a giant collapse value. □

For example, even a core $E = h_\alpha(x) + h_\beta(h_\gamma(y) + z)$ outside a proposed balanced certificate may be used in $3h_\lambda(E + c) - 2h_\mu(2E + d) - E$. Its support is still $(x, y, z)$. This result is a comparison theorem for that row, not a graph definition or a general entry-map algorithm for the core. An unrelated open term outside $E$, a variable ordinal head, or a nonlinear dependence on $E$ is outside the statement.

## 10 What these decisions do and do not establish

All source consumers begin by computing the literal canonical assignment. They then compare its closed positive coordinates and check the appropriate scalar certificate. This direction is essential: none asserts that every residual expression in its mathematical grammar is the image of a $T$ term. A supplied certificate may make a recognition procedure incomplete as a search for all possible decompositions, but its check and the ensuing comparison are exact on accepted inputs.

The finite-coefficient theorem includes all recursor-free ground-open sources at arbitrary internal type levels. The polynomial theorem includes all ground-open sources whose *entire* internal syntax has type level at most two, including recursors. At arbitrary internal levels, common-head, balanced, or single-core certificates can admit some nonpolynomial heads. Every result keeps repeated variables and correlated subexpressions, exact prefix masses, and all complete descent counters.

The mathematical algorithms are total; they are not promises of practical execution. A resource-limited implementation which cannot evaluate a closed exception or complete the inherited arithmetic procedure must report that unmet computation. A symbolic name for a closed value is sufficient where it cancels exactly, but is not a numerical evaluation when its sign is needed. We have neither implemented Snyatkov's full procedure here nor computed any literal giant $F_5$ value.

The remaining ground-open region includes unbalanced multivariate arrangements of arbitrary nonpolynomial heads, without any supplied certificate bringing them into the proved classes. The whole ground-open weak and strict comparison problem remains open. The type-confined ordering and non-ground expressivity companions address other parts and boundaries of the original effectiveness question [4, 6]; one-variable comparison has its own stronger methods [5]. None of these restricted results, individually or combined, announces a full resolution of the Wilken–Weiermann question.

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